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Principles of Programming
(WiSe 26/27)
Tutorial 1

Tasks marked with a star like **Optional Task*** are optional. Tasks marked like **Hard Task+** are given, but it is not expected that you solve them now. It is great if you learn to solve them during the lecture. Go back to them after a few weeks and see your own progress.

Learning Outcome: Basic Computing

1 Python Loops

Task 1: For Loops

Use one or more `for` loops to generate and print the multiplication table for the numbers 1 through 10.

- The rows should represent the multiplicands (1–10).
- The columns should represent the multipliers (1–10).
- Each cell should contain the product of the row and column values.
- Do not manually print each row. Generate the table using loops.

Expected output:

```
1   2  3  4  5  6  7  8  9  10
2   4  6  8 10 12 14 16 18  20
3   6  9 12 15 18 21 24 27  30
4   8 12 16 20 24 28 32 36  40
5  10 15 20 25 30 35 40 45  50
6  12 18 24 30 36 42 48 54  60
7  14 21 28 35 42 49 56 63  70
8  16 24 32 40 48 56 64 72  80
9  18 27 36 45 54 63 72 81  90
10 20 30 40 50 60 70 80 90 100
```

Task 2: While Loops

Write a function that takes two integer inputs:

- `base`
- `exponent`

Use a `while` loop to calculate the value of $base^{exponent}$ using repeated multiplication.

Do not use Python's built-in `pow()` function or the `**` operator.

Print the result in the format shown below.

Example
Expected output for input base=2 and exp=5:

```
2 to the power of 5 is:  32
```

2 Python List Comprehensions

Task 3: Reverse String

Using a list comprehension, create a function that returns the characters of a string in reverse order. The function should return a list containing the reversed characters.

Example
Expected output for input s = "hello":

```
["o","l","l","e","h"]
```

Note: Use a list comprehension rather than Python's built-in `reverse()` method or slicing syntax

Task 4: Sort Array

Given an integer array `nums`, create a new array in which all even numbers appear first, followed by all odd numbers. Use one or more list comprehensions to construct the result. The relative order of the even numbers and odd numbers does not matter.

Example
Expected output for input nums = [3,1,2,4]:

```
[2,4,3,1]
```

Explanation: The outputs [4,2,3,1], [2,4,1,3], and [4,2,1,3] would also be accepted.

Task 5: Fizz Buzz

Fizz buzz is a group word game for children to teach them about division. Players take turns to count incrementally, replacing any number divisible by three with the word "fizz", and any number divisible by five with the word "buzz", and any number divisible by both 3 and 5 with the word "fizzbuzz" (from Wikipedia).

Given an integer n, return a string array answer (1-indexed) where:

a. answer[i] == "FizzBuzz" if i is divisible by 3 and 5.

b. answer[i] == "Fizz" if i is divisible by 3.

c. answer[i] == "Buzz" if i is divisible by 5.

d. answer[i] == i (as a string) if none of the above conditions are true.

Example
Expected output for input n = 15:

```
["1","2","Fizz","4","Buzz","Fizz","7","8","Fizz","Buzz","11","Fizz
    ","13","14","FizzBuzz"]
```

Task 6: Numpy

Import the NumPy package (official documentation: https://numpy.org/doc/) and use it to solve the following tasks:

a. Print the NumPy version and configuration information.

b. Create a 10×10 matrix filled with zeros.

c. Determine and print the shape of an array (e.g., the matrix created in Task 1).

d. Create a 4×6 matrix filled with zeros and set the element at index `[2, 3]` to 1.

e. Create a 3×7 matrix containing consecutive integers starting from 21.

f. Create a vector of length 10 filled with random values between 0 and 1.

g. Reverse the vector from Task 6 (i.e., the first element becomes the last).

h. Create a 3×3 matrix containing random integer values between 0 and 7.

i. Find the indices of all non-zero elements in:

```
[1, 0, 7, 0, 1, 0, 0, 1, 0, 0, 1]
```

j. Find all unique values in:

```
[1, 0, 7, 0, 1, 0, 0, 1, 0, 0, 1]
```

k. Create a 3×3 identity matrix.

l. Multiply a 5×3 matrix by a 3×2 matrix.

m. Create a random vector of length 10 and sort it in ascending order.

n. Create a 5×5×5 array of random values and determine its minimum, maximum, and mean values.